

Tender evidence gap dossier - TR/REP&PM/ICB/2026/001/C

DERIVED GAP REGISTER — NOT A COMPLIANCE DETERMINATION. This dossier records, for each controlling clause, what the tender requires against what the bid pack was found to contain. It is an aid to querying the OEM and to closing evidence gaps before submission. It does not determine compliance, it does not bind the procuring authority, and it confers no waiver of any tender requirement.

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0. Document control

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Control field	Controlled value
Tender	TR/REP&PM/ICB/2026/001/C
Tender title	Establishment of 250 MW / 1000 MWh Standalone Battery Energy Storage System from 10 MW / 40 MWh AC Capacity Projects on Build, Own and Operate (BOO) Basis with 15 Year Operational Period
Addressed to	Envision Energy / the supplying group entity
Raised by	Hayleys
Submission deadline	4 September 2026, 10.00 hrs (Addendum No. 01 item 01)
Working days remaining	6
Open items	21 (5 critical, 6 high, 7 medium, 2 low, 1 informational)
Register digest	b8593e8c644898eb
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Status	Derived gap register - not a compliance determination

1. Critical path - raise these first

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These gate other work: nothing downstream can start until they are answered.

- 1. A1 - No artifact demonstrates grid-forming behaviour under fault at the required SCR
- 2. A2 - The RTE guarantee sits exactly on the floor on a basis that has since been hardened four ways
- 3. A5 - Grid Interconnection Confirmation Letter — a Major Deviation whose request window has closed

2. A1 - No artifact demonstrates grid-forming behaviour under fault at the required SCR

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Field	Content
Severity	CRITICAL
Controlling clause	Annex A A.05.17(i) and (j); Volume I §3(c); Annex A A.05.23(e)
Delivery tier	critical path
What the tender requires	Annex A A.05.17(i): 'The BESS shall support both grid-following and grid-forming control modes, enabling online switching between the two.' A.05.17(j): 'The grid-following mode shall operate stably under a minimum short-circuit ratio (SCR) of 1.2, while the grid-forming mode shall operate stably under a minimum SCR of 1.0.' Volume I §3(c) additionally requires a voltage-sourced, voltage-controlled inverter that 'should NOT change the control mode to current-controlled (grid-following) during normal operation or under any network fault conditions'.
What the pack contains	Three inconsistent characterisations of the same converter: (1) item 37, a signed letter asserting full grid-forming operation at all times; (2) item 61, a grid compliance list describing GFM and GFL modes with 'seamless transition' at clauses 3.17.4 and 3.17.4.4; (3) item 49, a PSCAD model that is the GFL variant, represented as a current source. A PSS(R)E grid-forming model with a Sri-Lanka dynamic record IS supplied, so the RMS grid-forming path is real.
Why that does not close it	A current-source EMT model cannot demonstrate voltage-source behaviour under fault, and the PSS(R)E GFM manual itself records that an EMT tool is the better recommendation for transient studies including fault ride-through. No supplied artifact therefore demonstrates grid-forming fault behaviour, and none addresses the SCR 1.0 grid-forming stability floor at A.05.17(j). IMPORTANT CORRECTION to the 31 July gap statement and to the 21 August review: the dual-mode capability itself is REQUIRED by A.05.17(i) and must NOT be withdrawn from the bid. A flat 'no grid-following capability' declaration would now be non-compliant.
Stated tender consequence	Annex A A.05.23(e) makes the grid-forming demonstration in both PSS(R)E and PSCAD a condition of the post-award Dynamic Model Tests. Addendum 01 items 06 and 12 make those results due within one month of ESA execution (14 January 2027) and add failure to deliver them acceptably to the grounds for forfeiting the Performance Security.

Field	Content
Question to Envision Energy / the supplying group entity	1. Does a grid-forming PSCAD/EMTDC model of the ENPCS2520 exist? If the GFL model was supplied in error, please issue the GFM EMT model. If no GFM EMT model exists, please say so plainly — that answer is needed now, not at ESA signature. 2. Please issue a single reconciled control-mode statement in these terms: the converter supports both modes with online switching as A.05.17(i) requires; it operates in grid-forming mode; it does not revert to current-controlled operation during normal operation or under network fault conditions as Volume I §3(c) requires; and grid-forming operation is stable to SCR 1.0 per A.05.17(j). 3. Please re-point the 'seamless GFL/GFM transition' language at clauses 3.17.4 / 3.17.4.4 so it is offered as the A.05.17(i) capability and NOT as the basis for converter robustness under fault.
Closes when	A grid-forming EMT model in PSCAD 5.x that initialises and runs at an SCR of 1.0, producing V/P/Q traces through a deep fault, with the controller remaining in voltage-source mode throughout; plus the reconciled written control-mode statement.

3. A2 - The RTE guarantee sits exactly on the floor on a basis that has since been hardened four ways

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Field	Content
Severity	CRITICAL
Controlling clause	Volume I §2.8(iii); clarifications 5, 31, 40, 51 and 55(b)
Delivery tier	critical path
What the tender requires	A guaranteed minimum AC-to-AC round-trip efficiency of 85 %, assessed monthly, with liquidated damages at 150 % of the peak-time 33 kV General Purpose tariff on the excess losses. Clarification 5: a monthly shortfall 'shall not be carried forward or reconciled' against later months or the year end. Clarification 51: 'Auxiliary and HVAC loads that are directly associated with the operation of the BESS Facility ... shall be accounted for in the RTE calculation.' Clarification 31: energy lost to frequency or voltage regulation in standby, unscheduled by the grid, is counted in RTE. Clarification 55(b): energy exchanged providing frequency response, synthetic inertia, oscillation damping and AGC is NOT excluded from the RTE calculation. Clarification 40, asked whether a 1 % test error could be treated as meeting the standard: 'No.'
What the pack contains	The 5 August revision of the 10 MW design calculation moves year-15 RTE including auxiliaries from 84.9 % to exactly 85.0 %, by changing the DC container generation from ENS-D10E to ENS-D10G, relaxing the MV cable efficiency assumption from 99.5 % to 99.6 %, and adding a clause deeming RTE compliant within an assumed 0.5 % electricity-meter error. Auxiliaries are computed at 35 degrees C. The document still carries version V1.0 with no revision record.
Why that does not close it	Clarification 40 refuses the measurement-tolerance principle outright, so the 0.5 % meter-tolerance clause has no contractual counterpart. Clarifications 31 and 55(b) put two loss terms INSIDE the measured quantity that the design calculation does not model at all. Clarification 51 closes off the aux-exclusive basis (88.4 %) that the 30 July review had hoped might apply. Clarification 5 removes annual averaging. A guarantee sitting exactly on the floor, on a basis carrying two unmodelled loss terms, assessed monthly with no reconciliation and no tolerance, is a structurally short position rather than a thin margin.
Stated tender consequence	Liquidated damages at 150 % of the peak-time 33 kV GP tariff on excess losses, assessed every month independently, with no annual reconciliation.

Field	Content
Question to Envision Energy / the supplying group entity	1. Please re-issue the guaranteed RTE schedule on the basis clarifications 51, 31 and 55(b) actually establish — auxiliary and HVAC inclusive, and including standby-regulation and ancillary-service energy — and WITHOUT the meter-tolerance clause, which clarification 40 has refused. 2. Please state the auxiliary load and the resulting guaranteed RTE at 45 degrees C ambient, not 35 degrees C. 3. If 85.0 % cannot be held on that basis in every year, please say so now and state the year and the margin, so the bid can be priced against it. 4. Please re-issue the 5 August document with a proper revision number and a change record against the 29 July version.
Closes when	A guaranteed monthly RTE schedule, by year, at 45 degrees C ambient, aux-inclusive, with the standby and ancillary-service terms itemised, showing positive margin above 85.0 % in every year with no tolerance clause relied upon.

4. A3 - Cell cycle life at tropical temperature is not substantiated against the contracted throughput

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Field	Content
Severity	CRITICAL
Controlling clause	Volume I §3.1(k) and §3.1(m); Annex A A.05; clarification 55(b)
Delivery tier	cannot close in window
What the tender requires	Volume I §3.1(k) requires demonstration of four hours at rated output at site ambient conditions at commissioning; the site envelope runs to +45 degrees C. The ESA contracts for 400 full equivalent cycles per year with a floor of 20 per month — about 6,000 cycles over the 15-year term as a MINIMUM obligation. Clarification 55(b) confirms that energy exchanged providing frequency response, synthetic inertia, oscillation damping and AGC is not excluded from that 400-cycle allowance.
What the pack contains	The supplier's own independent test-house bankability study (Issue D Final, 28 January 2026) records approximately 10,000 cycles to 70 % capacity retention at 25 degrees C and 0.25P, but only approximately 4,000 cycles at 45 degrees C — a 60 % reduction. The 20-year / 70 %-retention simulation the test house endorses is qualified to 25 degrees C, at most 50 % SOC and one cycle per day. Checklist items 55 and 56 nonetheless declare 'No Augmentation needed during lifetime' and 'No replacement needed during lifetime', and both status cells are blank.
Why that does not close it	At the 45 degrees C figure in the supplier's own independently reviewed data the cell reaches the 70 % threshold roughly 2,000 cycles short of what the ESA contracts for. Stated fairly: the system is liquid-cooled, so cell temperature is not ambient temperature, and 45 degrees C ambient-soak is not a prediction of cell temperature in service. That is precisely the problem — the package contains no thermal model, no cell-temperature substantiation at 45 degrees C ambient, and no auxiliary-load figure above 35 degrees C. The cooling-energy gap and the cycle-life gap compound, because the same missing thermal case sets both.
Stated tender consequence	This is the deepest exposure in the package for a 15-year BOO structure, and it is evidenced by the supplier's own independent report rather than by inference.

Field	Content
Question to Envision Energy / the supplying group entity	1. Please supply the thermal case at 45 degrees C ambient: coolant supply temperature, resulting steady-state CELL temperature at the 0.25P duty, and the cell-to-cell spread. 2. Please state the chiller capacity and electrical draw at that condition — the industry convention is an L45/W18 rating point, i.e. rated cooling output at 45 degrees C ambient with an 18 degrees C fluid supply. 3. At the resulting cell temperature, please state cycles to 70 % retention and reconcile that against the roughly 6,022 EFC the 15-year duty requires. 4. If the no-augmentation declaration cannot be substantiated at that cell temperature, please offer instead a capacity/throughput warranty with an augmentation undertaking triggered on measured capacity falling below the declared curve.
Closes when	A thermal calculation showing cell temperature at 45 degrees C ambient under the contracted duty, tied to a cycle-life figure at that cell temperature that exceeds 6,022 EFC with margin — or, failing that, a written augmentation undertaking with a defined measurement trigger and cost allocation.

5. A4 - Liquidated damages are uncapped over the term and the capacity charge has no monthly floor

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Field	Content
Severity	CRITICAL
Controlling clause	Volume III Appendix A Clause 2; clarifications 52 and 54; Volume I §7.1
Delivery tier	document
What the tender requires	Clarification 54: 'There is no aggregate cap on liquidated damages either per Contract Year or over the full 15-year Term.' The only cap is monthly. Further: 'capacity charge deductions for failure to achieve the required 97 % availability are not liquidated damages and therefore do not fall within the monthly LD cap', and 'if the BESS fails to meet the 97 % availability requirement, the capacity charge payable for that month may be reduced, potentially down to LKR 0'.
What the pack contains	No reliability block diagram, failure-rate data, redundancy analysis, MTTR, planned-maintenance schedule or spares policy has been supplied to support a 97 % monthly availability position. Operator reference letters report 98.2 % availability with a consistent fault signature — 0 hours cell-fault downtime, 36-43 hours DC-system, 68-70 hours PCS, 1 hour EMS.
Why that does not close it	This CORRECTS the 31 July gap statement, which recorded a 'total monthly LD cap of 20 % of capacity charge' and treated that as the bound on performance exposure. The availability deduction sits OUTSIDE that cap and has no floor, so a bad month can pay zero while damages continue to accrue with no term aggregate. Clarification 52 adds that no termination compensation or buy-out formula is prescribed for NSO default, political force majeure attributable to the Government, or prolonged natural force majeure. Volume I §7.1 and clarification 30 confirm the ESA is final and unamendable after the Letter of Award, so none of this is negotiable later. Note also that PCS is the dominant availability risk in the operator record, and the offered PCS model appears in none of the references (gap B4).
Stated tender consequence	Revenue can fall to zero in a month while damages accrue without a term cap.

Field	Content
Question to Envision Energy / the supplying group entity	1. Please provide a reliability block diagram, component failure rates, MTTR and spares holding sufficient to support 97 % monthly availability at the 95 % confidence level. 2. Please confirm a back-to-back availability warranty with defined response times, remote diagnostics and OEM-caused liquidated-damages indemnity — noting that the Project Company's exposure here is uncapped over the term and unfloored in any month. 3. Please state the spares list, location and replenishment lead time for the PCS specifically.
Closes when	A signed availability warranty with a service-level schedule and an OEM indemnity that responds to the uncapped, unfloored structure clarification 54 describes.

6. A5 - Grid Interconnection Confirmation Letter — a Major Deviation whose request window has closed

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Field	Content
Severity	CRITICAL
Controlling clause	Addendum 01 items 02, 07 and 08; Volume II Section 15; clarification 11
Delivery tier	critical path
What the tender requires	Addendum 01 item 02 adds the Grid Interconnection Confirmation Letter to the documents required with the proposal where Option 2 is selected under Volume II Section 11, and item 08 adds it to the Major Deviations list at Volume I clause 6.3.1. Clarification 11 confirms it 'is mandatory and shall form part of the Bid'. Addendum 01 item 07 requires it to be requested from the relevant Provincial Director of EDL 'at least 21 days before the Closing date', and states that EDL 'shall not be liable for any delays in issuance of the same'.
What the pack contains	Not evidenced in any material held in the corpus.
Why that does not close it	21 days before the 4 September closing date is 14 August 2026. That request window has passed. If Option 2 is being taken and the letter has not been requested, this is a disqualification risk rather than a scoring one. This gap is addressed to the Bidder, not to the OEM, and is recorded here because it outranks every technical item in the register.
Stated tender consequence	Major Deviation under Volume I clause 6.3.1 — the proposal may be rejected.
Question to Envision Energy / the supplying group entity	Bidder action, not an OEM query: confirm immediately whether Option 1 or Option 2 is being taken for the Grid Point; if Option 2, confirm whether the letter was requested and whether it will arrive before 4 September; if Option 1, record that election explicitly in the bid so the absence of the letter is not read as an omission.
Closes when	Either the executed Grid Interconnection Confirmation Letter in the bid, or a documented Option 1 election.

7. B1 - Annex A numeric performance parameters are not evidenced anywhere in the package

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Field	Content
Severity	HIGH
Controlling clause	Annex A A.05.02(a)(c)(d)(e), A.05.17(d) and (h)
Delivery tier	document
What the tender requires	A.05.02(a): PCS AC-side current continuous at 110 % of rated, sustaining 120 % for at least 2 minutes, preferably 150 % short-term. A.05.02(d): maximum inertia time constant 'no less than 20 seconds' with inertia response activation time at most 5 ms, flexibly adjustable. A.05.02(e): primary frequency regulation response under 0.2 s with active power adjustment deviation at most 2 %. A.05.17(d): autonomous suppression of 0.2-2.5 Hz oscillations with active power variation limited to 10-30 % Pn. A.05.17(h): AGC regulation range -100 % to +100 % Pn with steady-state active deviation at most 2 % Pn, and AVC steady-state reactive deviation at most 2 % Pn.
What the pack contains	The filled Volume 2 GTP declares droop 1-9 % and deadband 0.0-1.0 Hz in 0.05 Hz steps, which meet A.05.02(b). No other parameter in this list is evidenced. Clarifications 59 and 64 both asked for relief on the current ratings — including an explicit request to cut the 120 % duration from 2 minutes to 1 — and both were answered 'Please comply with A.05.02 of Annex A'.
Why that does not close it	These are hard numeric acceptance criteria in the controlling annex, and NSO has twice declined to relax the current ratings. The 110 % continuous requirement in particular implies roughly a 10 % PCS oversize on top of the reactive requirement, which is a sizing consequence and not merely a datasheet entry.
Question to Envision Energy / the supplying group entity	Please confirm, for the offered ENPCS2520 and at the ambient at which each holds: (a) continuous current at 110 % of rated, 120 % for at least 2 minutes, and whether 150 % short-term is supported and for how long; (b) inertia time constant and inertia response activation time, and the adjustment range; (c) primary frequency response time and active power adjustment deviation; (d) power-oscillation damping across 0.2-2.5 Hz and the active power variation band; (e) AGC and AVC regulation ranges and steady-state deviations.
Closes when	A datasheet or type-test extract giving each figure against the Annex A clause, at a stated ambient temperature.

8. B2 - The supplied Sri-Lanka protection envelope is narrower than Annex A requires at both ends

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Field	Content
Severity	HIGH
Controlling clause	Annex A A.05.04; clarification 29
Delivery tier	document

Field	Content
What the tender requires	Annex A A.05.04 sets steady-state operation across 47-52 Hz with extremes to 45-55 Hz, and an under-frequency window extending to 45 Hz with a 10-second requirement in the band $47.0 > f \geq 45.0$. Clarification 29, asked which of the two frequency ranges in the tender governs, answers: 'The Project Proponents shall comply with the requirements specified in Clause A 05.04 of Annex A.'
What the pack contains	The Sri-Lanka dynamic record ENVSG_PPC_2520_260416_LKA.dyr carries a 47.5 / 51.5 Hz continuous protection envelope with 46.9 / 52.1 Hz trip stages.
Why that does not close it	47.5 / 51.5 Hz is inside the 47-52 Hz steady-state band Annex A requires, so the plant as parameterised would trip within the range in which it is required to operate continuously — at both the under- and over-frequency ends.
Stated tender consequence	UNVERIFIED: the 47.5 / 51.5 Hz figures are carried forward from the 21 August ingress evaluation of the dossier, whose binaries are held outside the repository. They have not been re-read from the .dyr file for this register and must be confirmed against the delivered model before being put to the OEM as fact.
Verification	UNVERIFIED - treat as a question, not a finding
Question to Envision Energy / the supplying group entity	Please reconcile the .dyr protection envelope against Annex A A.05.04. Either confirm the settings are project-adjustable to at least 47-52 Hz continuous with the 45-55 Hz extremes and the 10-second requirement between 47.0 and 45.0 Hz, and state the adjustable range; or confirm the equipment limit if it cannot meet it.
Closes when	A revised dynamic record, or a written settings range, demonstrating continuous operation across 47-52 Hz and ride-through to the A.05.04 extremes.

9. B3 - Reactive capability may be marginally short if 11 MW is declared

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Field	Content
Severity	HIGH
Controlling clause	Annex A A.05.13; clarifications 2, 24 and 66
Delivery tier	document
What the tender requires	Clarification 24: 'The reactive power capability requirement shall be ± 0.3 times the rated active power of the BESS' — settling that it is a ratio, not the flat ± 3 Mvar referenced elsewhere. Clarification 2 confirms the Contracted Capacity will be the Declared Capacity offered in Section 1 of Volume II, expressly contemplating up to 11 MW. Clarification 66 confirms the declared capacity must be achieved and demonstrated at commissioning.
What the pack contains	The 10 MW / 40 MWh design calculation states ± 3.29 Mvar at the PCC with power factor ± 0.95 .

Field	Content
Why that does not close it	At a declared 10 MW the requirement is +/-3.0 Mvar and +/-3.29 Mvar passes comfortably. At a declared 11 MW the requirement becomes +/-3.3 Mvar, and +/-3.29 Mvar would fall marginally short. Since the 11 MW / 44 MWh variant carries more RTE headroom than the 10 MW case — which gap A2 makes valuable — the reactive figure is the constraint that decides whether the 11 MW declaration is available.
Stated tender consequence	UNVERIFIED: the +/-3.29 Mvar figure is stated in the 10 MW design calculation. The 11 MW design calculation's reactive figure has NOT been read for this register. The shortfall is therefore a flagged arithmetic risk to check, not an established finding.
Verification	UNVERIFIED - treat as a question, not a finding
Question to Envision Energy / the supplying group entity	Please state the guaranteed reactive capability at the PCC for the 11 MW / 44 MWh configuration, across the full voltage range 0.9-1.1 pu, both charge and discharge, at minimum and maximum SOC, at beginning and end of life, and at 45 degrees C. If it is +/-3.3 Mvar or better, please confirm it explicitly against clarification 24.
Closes when	A four-quadrant P-Q capability chart at the PCC for the offered configuration at the declared capacity, showing at least +/-0.3 x rated active power throughout.

10. B4 - The offered PCS model has no track record in the submitted reference list

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Field	Content
Severity	HIGH
Controlling clause	Volume I clause 2.7.3; clarification 58
Delivery tier	document
What the tender requires	Manufacturer qualification requires 1 GWh of installed battery modules/cells, 1 GW of installed inverter products, and two projects using the same PPC product family. Clarification 58(a) and (b) settle that the threshold is CUMULATIVE GLOBAL INSTALLED volume and must be installed/ commissioned volume, 'because the requirement says installed'. Clarification 58(e) settles that the qualification rests on the manufacturer's experience and 'need not be experience of the Project Proponent itself'.
What the pack contains	A supplier-experience workbook listing 22 PCS reference projects using ENPCS 2750, 3450, 3300 and 2500, and 24 battery-side projects. The Envision letter discloses 18.7 GW / 52.6 GWh contracted, 33.5 GWh shipped and 20.65 GWh at COD.
Why that does not close it	The model offered for Sri Lanka is the ENPCS2520, which appears nowhere in the reference list. The client-contact column the RFP form requests is empty for every PCS and PPC row, and the workbook is self-labelled '(part)'. Separately, clarification 58(b) means the qualifying figure is the 20.65 GWh at COD, not the 52.6 GWh contracted or 33.5 GWh shipped — the bid should quote the right number. This matters more than a paperwork point because PCS is the dominant availability risk in the operator reference record (68-70 hours of PCS downtime against 0 hours of cell-fault downtime).

Field	Content
Question to Envision Energy / the supplying group entity	1. Please explain the ENPCS2520's relationship to the referenced ENPCS 2750 / 2500 family and identify any commissioned ENPCS2520 units, with project, location and commissioning date. 2. Please complete the client-contact column for every PCS and PPC reference row. 3. Please issue the reference workbook in complete form rather than '(part)'.
Closes when	A manufacturer declaration on letterhead in the form clarification 58(c) accepts — product, installed quantity, project, location, commissioning date, manufacturer — covering the offered PCS model or a documented equivalence to the referenced family.

11. B5 - Fire-safety chain stops at module level and does not cover the offered container

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Field	Content
Severity	HIGH
Controlling clause	Volume I §3.1(g); UL 9540A 2026 (sixth edition)
Delivery tier	cannot close in window
What the tender requires	Volume I §3.1(g) requires fire prevention, detection, alarm, suppression, thermal-runaway mitigation, compartmentalisation, emergency isolation and safety procedures, with evidence submitted alongside the proposal.
What the pack contains	A cell-level UL 9540A (2019) report and a module-level UL 9540A (2026) report issued 25 June 2026, plus a Fire Protection System Specification V2.0 dated 21 May 2026. Recorded module-level results: peak heat release rate 46.28 kW, peak smoke release rate 0.5457 m ² /s, total smoke release 29.98 m ² , total hydrocarbons 432.6 L, module weight loss 1.8 kg, no flaming observed.
Why that does not close it	The module report's own conclusion records that cell-to-cell thermal runaway and propagation occurred, that runaway was contained by the module design, and that cell vent gas was flammable at cell level — and then states that a further test level is required. Neither further test is in the package. Separately the Fire Protection System Specification is headed for ENS-D10 while the offered configuration pairs ENS-D10G with ENS-D06G, so fire-protection coverage of the D06G variant is unevicenced.
Question to Envision Energy / the supplying group entity	1. Under the sixth edition of UL 9540A, published 13 March 2026, the unit-level test is no longer required for non-residential BESS and the installation-level large-scale fire test at Section 10 is the integrated fourth evaluation level. Please confirm which route applies to the offered product and supply a dated test commitment. 2. Please confirm fire-protection coverage of the ENS-D06G container specifically, or issue a specification revision whose header scopes both offered container types.
Closes when	Either an installation-level large-scale fire test report for the offered configuration, or a dated written commitment to it with a stated interim compliance basis; plus a fire-protection specification covering ENS-D06G.

12. B6 - Entity attribution and the missing Manufacturer's Authorization Letter

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Field	Content
Severity	HIGH
Controlling clause	Volume I clause 2.7.3; checklist item 58; clarification 58(d) and (e)
Delivery tier	document
What the tender requires	Checklist item 58 requires three signed EOI letters 'and MAL Also (on Company letterhead)'. Clarification 58(d): an OEM's standard EOI or letter format is acceptable provided it carries all RFP-required information and is signed by an authorised representative of the manufacturer.
What the pack contains	Three EOI letters (BESS; PCS and Transformers; SCADA/PPC and EMS), signed 31 July 2026 by an affiliated supply entity on the battery affiliate's letterhead. The clause 2.7.3 track record is in Envision Energy's name. The cell certificates name the affiliate's cell-manufacturing entity. Item 59.2 is a group letter recording that the BESS business has moved to a group battery affiliate.
Why that does not close it	Clarification 58(e) substantially relieves this — the qualification rests on the manufacturer's experience and need not be the Proponent's. What it does not resolve is WHICH group entity holds it, when the track record, the supply commitment and the certificates sit in three different corporate names. And no MAL is in the package at all, though item 58 is marked Received.
Question to Envision Energy / the supplying group entity	1. Please issue the Manufacturer's Authorization Letter on company letterhead. 2. Please confirm in writing that Envision Energy's clause 2.7.3 track record is available to, and may be relied upon by, the affiliated supply entity that signed the EOI letters, and supply any parent guarantee or intra-group support letter needed to make that reliance effective.
Closes when	A signed MAL, plus a written intra-group attribution confirmation naming the entities and the basis of reliance.

13. C1 - Fourteen standards remain uncertified, and the stated basis is stale in two places

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Field	Content
Severity	MEDIUM
Controlling clause	Volume I §3.1(b); clarification 62
Delivery tier	document
What the tender requires	Volume I §3.1(b) requires documentary proof of compliance for the listed battery, PCS, safety, quality and grid standards.

Field	Content
What the pack contains	Sixteen certificates and reports across checklist sections C and D. Still uncertified: IEC 62620, IEC 62902, IEC 62485-5, IEC 62933-1, IEC 62933-2-1, IEC 62933-5-2, UL 9540, IEEE 1547-2018, IEEE 2800-2022, UL 1741-SB and IEC TS 62786-3. IEEE 2800-2022 is declined outright as 'US region-specific ... not required for other markets'; UL 1741-SB is declined with EN 50549-2 and G99 offered instead. Two remarks read 'Will finish before 2026'.
Why that does not close it	The 'Will finish before 2026' remarks are stale — the date has passed — and a stale commitment reads worse than an honest revised one. The IEEE 2800 refusal is the substantive one: it is the transmission-level IBR interconnection standard whose grid-forming provisions are the natural reference for this tender's mandatory grid-forming requirement, and it is declined on market-scope grounds.
Question to Envision Energy / the supplying group entity	1. Please re-date the two 'Will finish before 2026' commitments with realistic completion dates. 2. Clarification 62 permits, where the exact package is newly introduced and certification is in progress, submission of currently valid certifications for the applicable cells, racks, major components or established product family TOGETHER WITH documentary evidence of the relationship and technical equivalence to the offered package. Please supply that equivalence evidence for each uncertified standard. 3. For IEEE 2800-2022 and UL 1741-SB, please state the equivalence argument clause-by-clause against EN 50549-2 and G99 rather than asserting market scope. Note that no published equivalence mapping between these standards exists, so the argument has to be constructed and will be read closely.
Closes when	For each uncertified standard: either a certificate for the offered package, or a clause-level equivalence statement with supporting certification for the established product family, in the form clarification 62 describes.

14. C2 - Item 57 answers a different requirement than the one it is filed against

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Field	Content
Severity	MEDIUM
Controlling clause	Volume I §3.1(m); checklist item 57
Delivery tier	document
What the tender requires	End-of-life decommissioning and battery RECYCLING commitments, referenced to Sri Lanka environmental regulations.
What the pack contains	A BESS dismantling technical description — a mechanical disassembly work instruction covering special tools, cable disconnection, grounding removal, pipeline removal and lifting procedure.
Why that does not close it	A keyword sweep of the full extract returns zero occurrences of recycling, disposal, waste, take-back, second-life or Sri Lanka environmental regulation. The single 'environmental' hit refers to removal of the environmental-control pipeline, i.e. the cooling circuit. The item's status cell is blank in the tracking workbook, consistent with it never having been assessed.
Question to Envision Energy / the supplying group entity	Please supply a genuine end-of-life decommissioning and recycling commitment: take-back or recycling route, the receiving facility and its jurisdiction, the treatment of the LFP cell chemistry, and an express reference to the Sri Lankan environmental regulations that apply. The disassembly manual is useful and should be retained, but filed against a different requirement.

Field	Content
Closes when	A signed recycling and decommissioning commitment citing the applicable Sri Lankan regulation.

15. C3 - Item 38 is filed as grid-forming modelling evidence but answers no tender requirement

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Field	Content
Severity	MEDIUM
Controlling clause	Checklist item 38; clarification 37
Delivery tier	document
What the tender requires	Checklist item 38 requires supporting documentary and MODELLING evidence of grid-forming operation.
What the pack contains	A file named as a grid-forming technical solution whose own title page identifies it as a BESS plant BLACK-START technical solution V1.0, marked 'for reference'. Its content is a PPC/SCADA functional description — topology monitoring, anti-misoperation interlocks, GNSS-synchronised zero-voltage ramp-up, off-grid operation, SOC balancing, load management, resynchronisation.
Why that does not close it	It contains no simulation results, no V/F characteristic, no fault response and no SCR sweep, so it is not modelling evidence. And clarification 37, asked whether the black start function is mandatory in this project, answers 'No' — so the document does not answer item 38 and is not required by the tender in its own right either.
Question to Envision Energy / the supplying group entity	Please replace item 38 with actual grid-forming modelling evidence — see gap A1. The black-start document may be withdrawn or retained as supporting information, but it should not stand against item 38.
Closes when	Simulation output demonstrating grid-forming V/F behaviour, filed against item 38.

16. C4 - Filled Volume 2 GTP contains unforced data-entry errors in a scored schedule

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Field	Content
Severity	MEDIUM
Controlling clause	Volume II Section 6
Delivery tier	document

Field	Content
What the tender requires	The completed Guaranteed Technical Particulars schedule.
What the pack contains	The completed Section 6 GTP, confirming the tender number. Substantive entries are sound: ENPCS2520 at 2.52 MW / 690 V / 50 Hz, liquid cooling, 416S1P x 6, 0.25C continuous, 100 % DoD, 7300 EFC at 100 % DoD / 9125 at 80 % / 14600 at 50 %, at most 3 %/month self-discharge, IP65, droop 1-9 %, deadband 0.0-1.0 Hz, PF +/-0.95.
Why that does not close it	A.5 'Model No.' is answered '8'; A.4 'Make' is answered 'BESS'; A.6 'Total Area Required (Acres)' is answered '350 m2' — wrong unit and implausible for the block. Block totals (A.15 12 MW, A.16 48 MWh, 8 modules, 8 inverters) reconcile to neither the 10 MW / 40 MWh nor the 11 MW / 44 MWh design calculation, and 8 x ENPCS2520 is 20.16 MW against a declared 12 MW battery rating. B.40 'Grid Forming Capability' is answered with a bare 'Yes'. The GTP carries no round-trip-efficiency row, no capacity-warranty row, no degradation row and no augmentation row, so the commercially load-bearing guarantees live only in the design calculations.
Question to Envision Energy / the supplying group entity	Please correct A.4, A.5 and A.6; reconcile the A.15/A.16 block totals to the offered configuration; expand B.40 to the reconciled control-mode statement at gap A1; and confirm whether the RTE, capacity-warranty, degradation and augmentation guarantees can be carried in the schedule rather than only in the design calculation.
Closes when	A corrected, internally consistent GTP that reconciles to one offered configuration.

17. C5 - Grid-code compliance monitoring is now unambiguously the Developer's scope

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Field	Content
Severity	MEDIUM
Controlling clause	Addendum 01 item 03; clarifications 4(b), 21 and 63
Delivery tier	document
What the tender requires	Clarification 21: NSO will NOT specify the make and model of the grid-code compliance monitoring equipment ('No'); the DEVELOPER is responsible for its ongoing maintenance and calibration; and the data from it WILL be used for billing and verification ('Yes'). Clarification 4(b) sets the standard as IEC 61000-4-30 Class A. Clarification 63 requires a separate firewall at the BESS facility with a suitable management system.
What the pack contains	In the 21 August package, Section E of the GTP answered 'Meter is Bop scope' on all five grid-code compliance metering rows, and items 43 and 44 were likewise declined as BoP scope.
Why that does not close it	The 21 August evaluation recorded that no party in the package owned grid-code compliance monitoring. That is now resolved against the Developer, and the monitoring data is billing-relevant, so a BoP-scope answer is no longer available. An IEC 61000-4-30 Class A instrument and a separate managed firewall are now specified items to be priced and supplied.
Question to Envision Energy / the supplying group entity	Please confirm whether the offered EMS/PPC package includes an IEC 61000-4-30 Class A power-quality instrument and the separate BESS-side firewall, or whether these remain outside OEM scope — and if outside, please say so explicitly so the Bidder can price them into BoP.

Field	Content
Closes when	A written scope statement covering the Class A instrument, its calibration regime and the BESS-side firewall.

18. C6 - SCADA gateway must serve four master servers without additional licences

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Field	Content
Severity	MEDIUM
Controlling clause	Addendum 01 item 09; clarification 27
Delivery tier	document
What the tender requires	The SCADA gateway 'shall support simultaneous communication and real-time data reporting to a minimum of four SCADA master servers, comprising two master servers at the National System Control Centre and two master servers at the Backup National System Control Centre', capable of reporting to any one, any combination, or all four 'without interrupting the existing communication links or requiring additional hardware, software, configuration changes, or licenses'. Clarification 27 confirms a gateway is still required even where the PPC itself supports IEC 104.
What the pack contains	Not evidenced. The package includes an EnOS PPC but no gateway specification against this requirement.
Why that does not close it	This is a new requirement introduced by Addendum 01 and post-dates the OEM material in the corpus. The 'no additional licenses' wording is a commercial constraint as well as a technical one.
Question to Envision Energy / the supplying group entity	Please confirm the offered SCADA gateway supports four simultaneous IEC 104 master connections in any combination, without additional hardware, software, configuration change or licence cost, and state the product and its concurrent session limit.
Closes when	A gateway datasheet stating concurrent master-server support and the licensing position.

19. C7 - Tracking workbook cannot be relied on as a completeness measure

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Field	Content
Severity	MEDIUM
Controlling clause	Bidder document control
Delivery tier	document

Field	Content
What the tender requires	An accurate internal record of which checklist items are satisfied.
What the pack contains	The supplier document checklist workbook, whose Summary tab uses COUNTIF over F2:F65 while the checklist data runs to row 69.
Why that does not close it	The Summary tab reports 38 Received against a true 41, because items 60, 61 and 62 fall outside the counted range. Separately, items 50-53 do not exist in the workbook at all — the numbering jumps from 49 (PSCAD model) straight to 54. Given the section they sit in, these are plausibly the SCR/phase-step and model-validation rows. Item 36 is also marked Not Received with no remark, although item 61 post-dates and largely answers it.
Question to Envision Energy / the supplying group entity	Bidder action, not an OEM query: correct the Summary range to F2:F69, restore items 50-53, set statuses for items 55-57, and update item 36 against item 61.
Closes when	A workbook whose Summary reconciles to the checklist rows and contains no gaps in numbering.

20. D1 - The clarification register contradicts itself on shared grid interconnection

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Field	Content
Severity	LOW
Controlling clause	Clarifications 20, 46 and 75
Delivery tier	document
What the tender requires	Clarification 20, on two projects at one GSS on a single land plot: 'Yes. A common grid interconnection line may be used, subject to compliance with the applicable technical requirements and obtaining the necessary approvals from the EDL.' Clarification 75, on the same question: 'Due to network constraints, only one BESS project can be connected to a single 33 kV feeder ... it is technically not feasible to accommodate both projects through the same feeder.'
What the pack contains	Not applicable — this is a defect in the controlling document, not in the bid.
Why that does not close it	The two answers point opposite ways on the same commercial question, and it determines whether a two-project bid at one GSS shares interconnection capital cost. They reconcile only if a common line can terminate in two separate feeder bays, which clarification 46's premise makes plausible but neither answer states. The clarification window closed on 25 August, so this can no longer be asked.
Question to Envision Energy / the supplying group entity	Bidder action: cost any two-project bid at one GSS on the conservative reading — separate 33 kV feeders per clarification 75 — and state that assumption explicitly in the bid so it is visible to the evaluator rather than discovered later.
Closes when	A stated, priced assumption in the bid recording which reading was taken and why.

21. D2 - Capacity Charge Rate unit is internally inconsistent within its own clarification

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Field	Content
Severity	LOW
Controlling clause	Clarification 65; Addendum 01 items 11 and 13
Delivery tier	document
What the tender requires	Clarification 65 answers the unit question with 'LKR/MW/month' and states 'Capacity Charge Rate proposed by the Project Proponent = Y LKR/MW/month', then gives 'Applicable Capacity Charge Rate: $0.15 \times Y + (0.85 \times Y \times P2/P1)$ LKR/MWh/month' — a different unit in the same answer, in the item whose whole purpose was to resolve unit ambiguity.
What the pack contains	Not applicable — a defect in the controlling document.
Why that does not close it	The proposed rate and the applicable rate cannot be in different units and remain the same quantity. Addendum 01 item 11 also removes '(excluding VAT)' from clause 2.8 and item 13 adds explicit SSCL and VAT-18 % rows to the Section 4 table, so the quoted figure's tax basis has changed as well.
Question to Envision Energy / the supplying group entity	Bidder action: quote in LKR/MW/month, which is the unit the answer opens with and the unit of the proposed rate, and state that election on the form. Complete the new SSCL and VAT rows per Addendum item 13.
Closes when	A Section 4 form completed in LKR/MW/month with the SSCL and VAT rows filled.

22. D3 - Two technical proposals are now permitted — the only remaining hedge

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Field	Content
Severity	INFORMATIONAL
Controlling clause	Addendum 01 item 23; clarification 73
Delivery tier	document
What the tender requires	Addendum 01 item 23 repeals the Volume I clause 2.7.1 note barring variant proposals and replaces it with: 'The Project Proponent may submit a maximum two (02) Technical Proposals under the Financial Proposal submitted for this RFP.' Clarification 73(a) confirms up to two alternative manufacturers or complete solutions, each 'a complete and technically integrated BESS solution', treated as 'separate and complete solutions', with NO interchange of major equipment between them during detailed design, construction or implementation. Clarification 73(b), asked whether a nominated supplier may be changed after award: 'No.'

Field	Content
What the pack contains	One configuration, with an unresolved question over which design calculation is the offered one.
Why that does not close it	This is an opportunity rather than a deficiency, recorded here because it is the principal change to bid strategy and it expires at submission. It maps directly onto the two unresolved technical exposures — the grid-forming EMT gap at A1 and the PCS track-record gap at B4 — and it is the ONLY hedge available, because supplier substitution after award is refused. It also supersedes the 31 July gap statement's repeated instruction that the RFP permits only one technical solution.
Question to Envision Energy / the supplying group entity	Bidder decision, informed by Envision's answer to A1: whether the second technical proposal should be the 11 MW / 44 MWh variant (more RTE headroom — see A2 and B3), a different PCS within the Envision range with an established track record (see B4), or a different OEM entirely.
Closes when	Two complete, internally consistent technical proposals, with no shared or interchanged major equipment.

23. Evidence inventory - declared against received

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The two columns are deliberately separate. Where they diverge, the divergence is the finding: a row marked received against a document that does not answer the requirement is a gap, not closure.

Item	Supplier declares	Audit found	Adequate?	Note
PSS(R)E RMS model, grid-forming	Checklist item 48 — Received	Executable DLLs, Sri-Lanka dynamic record ENVSG_PPC_2520_260416_LKA.dyr (ENGFM01 + BNPPC_GFMV3), manual V1.4a	adequate	Satisfies half of Annex A A.05.23(d)(I). The manual itself states EMT is the better tool for fault ride-through.
PSCAD/EMTDC EMT model	Checklist item 49 — Received	PCS2520x4_UPPC_x64_260605aBB.pscx — manual titled for the GFL PCS variant, model block labelled GFL-PCS, converter represented as a current source	NOT adequate	Accepted for the A. 05.23(d)(I) bid-stage minimum as an EMT model in PSCAD format, but cannot demonstrate voltage-source behaviour under fault.
Grid-forming capability letter	Checklist item 37 — Received	Signed Envision letter asserting full grid-forming voltage-source operation under all operating conditions	NOT adequate	Contradicted by the item 61 compliance list (dual-mode) and by the GFL EMT model. See gap A1.

Item	Supplier declares	Audit found	Adequate?	Note
Grid-forming modelling evidence	Checklist item 38 — Received	A BESS plant BLACK-START technical solution, V1.0, marked 'for reference'	NOT adequate	Not modelling evidence: no simulation results, no V/F characteristic, no fault response, no SCR sweep. Clarification 37 also confirms black start is NOT mandatory, so this document answers no tender requirement at all.
End-of-life recycling commitment	Checklist item 57 — filed	A mechanical disassembly work instruction (tool list, cable disconnection, lifting procedure)	NOT adequate	Zero occurrences of recycling, disposal, waste, take-back, second-life or Sri Lanka environmental regulation in the full extract.
Manufacturer's Authorization Letter	Checklist item 58 — Received	Three EOI letters present; NO MAL in the package	NOT adequate	-
Cell certification (UL 1973)	Certified	UL 1973 component recognition (BBGA2), cell level, in the cell-manufacturing entity's name	NOT adequate	The certificate states UL Recognized components are incomplete in certain constructional features. System-level UL 9540 is self-declared 'partially comply / future'.
Fire safety — UL 9540A	Checklist item 45 — Received	Cell-level (2019) and module-level (2026) reports; Fire Protection System Specification V2.0 headed for ENS-D10	NOT adequate	The module report's own conclusion requires a further test level. The offered configuration pairs ENS-D10G with ENS-D06G; D06G coverage is unevidenced.
Offered PCS track record	22 PCS reference projects supplied	References list ENPCS 2750, 3450, 3300 and 2500. The offered ENPCS2520 appears in none of them. Client-contact column empty on every PCS and PPC row	NOT adequate	-
Independent cell bankability study	Checklist item 47 — Received	Genuine third-party test-house Technical Advisory Report, Issue D Final, 28 January 2026, prepared for the battery affiliate	adequate	Adequate as an independent study — and it is the source of the 45 degrees C cycle-life exposure at gap A3.

24. What changed since the 31 July gap statement

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This register supersedes the 31 July 2026 detailed gap statement in four respects, each because a controlling document has since been obtained rather than because the earlier analysis was careless. (1) GRID-FORMING. The 31 July statement required a 'no-GFL-reversion statement' and the 21 August review went further, instructing deletion of the supplier's 'seamless GFL/GFM transition' language. Annex A A.05.17(i) REQUIRES support for both modes with online switching. Both instructions are withdrawn: acting on them would have removed evidence of compliance with a mandatory clause. The correct position is capability for both modes, operation in grid-forming, no reversion to current-controlled operation in normal or fault conditions, and grid-forming stability to SCR 1.0. (2) DYNAMIC MODELS. The 31 July statement read Volume I §3.1(p) correctly as models OR test results, and recommended supplying both models anyway. Annex A A.05.23(d) confirms that either/or, and Addendum 01 item 14 writes it into the Volume II compliance schedule. The bidder holds both models, so the stated rejection ground is clearable at bid stage; the SCR sweep moves to the post-award Dynamic Model Tests, due within one month of ESA execution with the Performance Security at risk. (3) COMMERCIAL EXPOSURE. The 31 July statement recorded a total monthly liquidated-damages cap of 20 % of the capacity charge. Clarification 54 establishes that there is no aggregate cap over the Contract Year or the Term, and that availability deductions sit OUTSIDE the monthly cap and can take the capacity charge to LKR 0. The exposure is materially larger than recorded. (4) BID STRATEGY. The 31 July statement repeatedly noted that the RFP permits only one technical solution. Addendum 01 item 23 now permits two.

25. Questions the 31 July statement recommended, and how the register answered them

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The 31 July statement listed ten clarification questions for NSO. The issued register answers or partly answers seven of them: Q2, RTE metering boundary — ANSWERED by clarification 51: auxiliary and HVAC loads directly associated with BESS operation are inside the RTE calculation; general site loads such as CCTV, security and lighting are outside it. Q3, dynamic evidence route — ANSWERED by Annex A A.05.23(d) and clarification 35: PSS(R)E and PSCAD are compulsory; PowerFactory, the third tool named at A.05.23(b), is not. Q6, black start — ANSWERED by clarification 37: not mandatory. Q7, P-Q envelope — PARTLY, by clarifications 24 and 32. Q8, capacity test timing — PARTLY, by clarifications 25, 26 and 66. Q9, annex package — PARTLY: Annex A, Addendum 01 and the clarification register are now held. ANNEXES B, C AND D REMAIN OUTSTANDING and are referenced by clauses this register relies on. Q10, export limit — REFRAMED by clarification 2: the

Contracted Capacity becomes the Declared Capacity, expressly up to 11 MW, so 11 MW is a declarable capacity rather than only a tolerance above 10 MW. Three were NOT answered and can no longer be asked, because the clarification window closed on 25 August 2026: Q1 (whether the Volume I 85 % monthly minimum or the Volume III FEC-banded schedule governs), Q4 (whether both phase-step polarities are required — Annex A A.05.23(d)(II) states +50 degrees, so the 31 July recommendation to test both polarities and record +50 as the explicit minimum stands), and Q5 (grid-forming fault current magnitude and acceptable current-limiting behaviour).

26. Closure pathways — how these gaps can be closed

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A1, GRID-FORMING AT SCR 1.0. The control approach matters here, not just the model. Power Synchronisation Control, which uses active power rather than a frequency measurement as the synchronising signal, is reported as the most stable grid-forming approach at very low short-circuit ratios below about 1.5, where PLL-based grid-following control approaches its critical SCR of 1.0. If the ENPCS2520 grid-forming firmware uses a virtual-synchronous-generator approach that reacts to frequency and ROCOF, its stability at SCR 1.0 should be demonstrated rather than assumed. For the validation package itself, IEEE 2800.2 is the companion guide covering model validation, verification and test procedures, and the published EMT model requirements of SPP, ERCOT and AEMO are usable templates for what an acceptable submission contains — they are the same deliverable NSO is asking for. Note that IEEE has issued an amendment, IEEE 2800a, specifically to reduce barriers for IBRs with grid-forming equipment; it is the right reference to cite if Envision argues grid-forming compliance outside the US market. A2 and A3, RTE AND CELL TEMPERATURE. These are one thermal problem with two contractual faces, and the closure route is the same calculation. Published practice is that a containerised system's HVAC load can move system RTE by 2-4 percentage points seasonally, and that cooling load at 40 degrees C ambient can consume 3-5 % of throughput — which is the order of the entire margin at issue in A2. The RTE test is run at rated power, where losses and auxiliary draw are both at maximum and auxiliary draw is dominated by battery cooling. The favourable side is that liquid cooling holds cell-to-cell spread below about 5 degrees C and, at a 0.5C duty, keeps module temperatures in a 20-30 degrees C band; the contracted duty here is 0.25P, gentler still. So the physics plausibly supports Envision's position — but only if the calculation is produced. The specifiable rating point is the L45/W18 convention: rated cooling output at 45 degrees C ambient with an 18 degrees C fluid supply. General industry data corroborates the bankability study independently: roughly every 10 degrees C of additional cell temperature doubles degradation rate, and sustained 45 degrees C versus 25 degrees C operation is reported to cut cycle life by 50 % or more, which is the same order as the study's 10,000 to 4,000 figures. B5, FIRE. The sixth edition of UL 9540A, published 13 March 2026, restructured this. The unit-level test is no longer required for non-residential BESS, and the installation-level large-scale fire test at Section 10 is now fully integrated as the fourth evaluation level after cell, module and

unit, aligned to NFPA 855 guidance and intended to demonstrate that fire will not propagate between ESS units. For a utility-scale installation the realistic route is therefore the Section 10 installation-level test. It is physical burn testing and cannot be completed before 4 September, so the deliverable for this bid is a dated commitment plus an interim compliance basis – which is a materially stronger position than silence, given that the module-level report itself tells the evaluator a further test is required. C1, STANDARDS. Clarification 62 is the route: currently valid certifications for the applicable cells, racks, major components or established product family, plus documentary evidence of the relationship and technical equivalence to the offered package. This is an evidence route, not a waiver. One caution stated plainly: no published equivalence mapping between IEEE 2800-2022 or UL 1741-SB and EN 50549-2 or G99 was found, so that argument must be constructed clause by clause and should not be presented as though a recognised equivalence already exists.

27. References for the closure pathways

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Grid-forming control and validation: 'Review of recent developments in grid codes: focus on compliance testing and grid-forming inverter-based resources', ScienceDirect, <https://www.sciencedirect.com/science/article/pii/S1364032125011827> . SPP Electromagnetic Transient (EMT) Model Requirements, https://opsportal.spp.org/documents/studies/SPP%20EMT_Model_Requirements_R1.pdf . MISO draft GFM BESS performance requirements whitepaper, <https://cdn.misoenergy.org/20240903%20IPWG%20Item%2004a%20DRAFT%20GFM%20BESS%20Performance%20Requirements%20Whitepaper.pdf> . IEEE 2800a, 'Reduce Barriers for IBRs with Grid-Forming Equipment', <https://standards.ieee.org/ieee/2800a/12386/> . Elia INPOWEL EMT models for grid-forming assets, <https://innovation.eliagroup.eu/en/projects/inpowel-emt-models-for-grid-forming-assets> . NREL, 'Review of Technical Requirements for Inverter-Based Resources', <https://docs.nrel.gov/docs/fy25osti/91792.pdf> . UL 9540A sixth edition: Intertek, 'Advancing Fire Safety in Energy Storage: Understanding the 2026 Update to UL 9540A', <https://www.intertek.com/blog/2026/03-23-understanding-the-2026-update-to-ul-9540a/> . Energy-Storage.News, 'UL9540A: shift to system-level testing defines new edition', <https://www.energy-storage.news/ul9540a-shift-to-system-level-testing-defines-new-edition-of-key-bess-safety-standard/> . Mayfield Renewables, 'The 6th Edition of UL 9540A is Here', <https://www.mayfield.energy/technical-articles/the-6th-edition-of-ul-9540a-is-here/> . UL Solutions, UL 9540A test method, <https://www.ul.com/services/ul-9540a-test-method> . Thermal and cycle life: Patsnap, 'LFP Battery Operating Temperature Guide', <https://www.patsnap.com/resources/blog/articles/lfp-battery-longevity-optimal-temp-ranges/> . SunLith Energy, '0.5C vs 1C cycle life in liquid-cooled BESS (LFP data)', <https://sunlithenergy.com/liquid-cooled-bess-0-5c-vs-1c-cycle-life/> . Bonnen Batteries, 'Liquid cooling for BESS', <https://www.bonnenbatteries.com/liquid-cooling-for-battery-energy-storage-systems-how-ci-bess-manages-heat/> . RTE and auxiliary load: Merus Power, 'Round-trip efficiency as a performance guarantee for BESS', <https://meruspower.com/blog/round-trip-efficiency-as-a-performance-guarantee-for-bess/>

formance-guarantee-for-battery-energy-storage-systems/ . Cooltechx, 'BESS liquid cooling capacity: why L45/W18 degrees C test conditions matter', <https://www.cooltechx.com/bess-liquid-cooling-capacity-l45-w18/> . These references support the CLOSURE PATHWAYS only. No reference above is evidence about the offered Envision product, and none of them is a tender document. Every requirement in the register above is anchored to the NSO documents listed under Source provenance.

28. Source provenance

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Every document read to build this register, with the route by which its text was recovered. A clause quoted from an OCR-recovered scan is not the same evidence as one quoted from a digital original.

Document	Role	Dated	SHA-256	Extraction	Note
NSO RFP Volume I — Instructions to Project Proponents	controlling tender volume	July 2026	fb61a4f827a0	native	-
NSO RFP Volume II — proposal forms and compliance schedule	controlling tender volume	July 2026	41644c207213	native	-
NSO RFP Volume III — Model Energy Storage Agreement	controlling tender volume	July 2026	d3413e9d1a1b	native	-
Addendum No. 01	controlling amendment — overrides the volumes it amends	7 August 2026	2b99b1e507d8	markdown	23 items. Complete embedded text layer; Attachments 1-4 are separate files and are NOT in this PDF or in the corpus.
Annex A — Functional and Performance Requirement	controlling technical annex	26 June 2026 (file creation)	be599073f597	markdown	27 pp. A.05.01(a) gives this annex precedence over the Grid Connection Code where they differ. Figures on 6 pages are not reproduced in the text extract.

Document	Role	Dated	SHA-256	Extraction	Note
Clarifications for the RFP — 76 numbered items	controlling clarification register	21 August 2026 (scan date)	73fbaca1a7e7	ocr	IMAGE-ONLY SCAN — no text layer. Markdown returned an empty document and tesseract scrambled the two-column table, so all 15 content pages were read as page images. Quotations here come from that verified transcript, not from the OCR text.
Envision design calculation, 10 MW / 40 MWh, V1.0	OEM evidence — superseded	29 July 2026	7281d964654e	native	Superseded by a 5 August document that still carries version V1.0.
Envision design calculation, 11 MW / 44 MWh, V1.0	OEM evidence — distinct candidate configuration	5 August 2026	0cf77ec5d761	native	Does not state the tender number.
Envision functional-requirements checklist	supplier declaration — not evidence	21 July 2026 (metadata)	8236806c21f6	ooxml	48 Yes / 7 No / 1 Partial across 56 rows. No tender identifier, signatory, revision or evidence-reference column.
DutchBay detailed gap statement (prior issue)	derived analysis — superseded in part by this register	31 July 2026	ce4a996893b5	markdown	-
Bidder evidence dossier, checklist sections A-J	OEM evidence — held outside the repository, manifest only	21 August 2026	not hashed	manual	72 files / 58 unique. Binaries are NOT committed (third-party copyright, compiled control code, personal data). Findings against this package are carried forward from the 21 August ingress evaluation, which read the originals.

29. Verification discipline

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Verification discipline: every gap below is anchored to a controlling clause and to the document actually received. A supplier declaration is recorded as a declaration and is never counted as evidence — a checklist row marked 'Received' against a document that does not answer the requirement is reported as a gap, not as closure. Where a finding could not be verified from the supplied material it is marked UNVERIFIED and its basis is stated, rather than being asserted or dropped.

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PDF variant pdf/ua-1; house stylesheet applied: True.
Font substitution: none; house fonts embedded: UNVERIFIED (poppler unavailable).
Gap register digest (SHA-256, first 32): b8593e8c644898eb7318450dfab03887.
Register content: 21 gaps, 10 evidence rows, 11 source documents.